Aditya Narayan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.43
Base PU College	Class XII (KSEAB)	2024	93.00%
AECS Magnolia Maaruti Public School	Class X (CBSE)	2022	95.20%

SCHOLASTIC ACHIEVEMENTS

Secured a **Top 10** finish in **Optiver Tradathon**, competing against students from multiple IITs.

Offered a seat in the **Indian Statistical Institute**.

Certification of Achievement received from Tata Capital.

PROJECTS

Algorithmic Strategy Development on **Multi-Feature Time Series**

- Represented IIT Hyderabad as a part of Inter IIT Tech Meet 14.0 hosted by IIT Patna.
- Explored and evaluated multiple approaches including XG-Boost variants, Fourier Transform based methods and different transformer models before converging on **modified VIDYA** (Variable Index Dynamic Average) indicator complemented by **statistically driven** trading strategies.
- Achieved a Sharpe Ratio of **2.61** and a Calmar Ratio of **2.7** on provided dataset.

Locality Sensitive Hashing (LSH) | *DSA Lab Course Project* | GitHub Link

- Engineered an efficient LSH indexing system for high-dimensional data, providing a high-speed stochastic alternative to deterministic Nearest Neighbor (NN) search algorithms.
- Achieved a huge reduction in query response time (accelerating searches from **~8.3ms to ~79µs**) compared to brute-force methods by partitioning data into hash bins and drastically reducing the average number of candidates checked per query.
- Benchmarked speed and accuracy across multiple hyperparameters (hash length and table count), and configured a model to achieve **76% recall** and a **1.01 approximation ratio** while maintaining **sub-2ms** query times.

Image Restoration & Enhancement using Transformers | KLA AI Hackathon | GitHub Link

- Built and trained a custom **Swin Vision Transformer** (SwinIR) in **PyTorch**, leveraging local window-based **multi-head self-attention** and PixelShuffle up-sampling to reconstruct heavily degraded, low-resolution inputs into 256x256 resolution images.
- Formulated a custom frequency-spatial composite loss function to enforce perceptual and structural accuracy, integrating Fast Fourier Transform (FFT) , SSIM, and L1 spatial penalties.
- Engineered a **stochastic data augmentation** pipeline incorporating CutMix, MixUp, and dynamic patch-masking to maximize model generalization against speckled noise and motion blur, establishing a strong baseline of **0.762 SSIM** and **25.83 PSNR** on given data.

LLVM IR Analysis & Lexer Implementation | Compilers Course Project

- Built a lexical analyzer in Flex for tokenizing C source code by recognizing identifiers, keywords, numeric literals, operators, comments, and delimiters with longest-match and rule-priority semantics.
- Explored the LLVM/Clang 17 compiler toolchain by generating and analyzing LLVM IR from C/C++ programs to study control flow, memory layout and data structures.
- Studied compiler optimization passes and IR transformations to understand control flow graphs, memory operations, and optimization techniques.

SKILLS

Programming Languages: C, C++, Python, SQL

Libraries & Frameworks: NumPy, Pandas, Matplotlib, SciPy, scikit-learn, PyTorch, Transformers.

Tools: Git, Github, Jupyter Notebook, LaTeX, MySQL

RELEVANT COURSES

Computer Science: Data Structures and Applications (along with Lab), Algorithms, Introduction to Programming, Compilers, Database Management System, Theory of Computation

Mathematics: Linear Algebra, Probability, Applied Statistics, Ordinary Differential Equations, Introduction to Number Theory, Complex Variables, Introduction to Group Theory, Calculus I & II

POSITIONS OF RESPONSIBILITY

Core at Epoch, the **AI-ML & Data Science Club** of IIT Hyderabad.

- Delivered multiple core machine learning sessions to the IITH community, engaging over 100 students per session.
- Actively participated in internal club hackathons, promoting hands-on learning and problem solving in AI/ML.

Teaching Assistant of MA1230 - Series of Functions

Head Internship Co-ordinator at OCS, the Career Cell at IIT Hyderabad

Finance Co-ordinator at FCC, the Finance and Consulting Club of IIT Hyderabad

- Authored and published a bi-weekly finance newsletter distributed to the entire IIT Hyderabad (IITH) community for one academic year.
- Delivered multiple sessions on basics of finance and fundamental analysis engaging over 60 students per session.

Core at Infero, the **Competitive Programming Club** of IIT Hyderbad